TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Laser-based digital thin-film manufacturing technology for metallic, ceramic and semi-conductor functional layers			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (x), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical Materials s		*Mechanical Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Surface Treatment	*Coating/ Method Technologies*	*Nano/Micro Mechanical Systems*	*Miniature Processing/ Assembly/Measurement Technologies*
Concept of Proposed Technology	The proposed technology aims at the development of a laser-induced forward transfer (LIFT) based digital manufacturing process for the deposition of highly functional metallic and ceramic thin films. The LIFT process enables maskless, contact-free, and digitally controllable material transfer with high spatial resolution. In contrast to conventional printing or vacuum-based coating technologies, LIFT allows the deposition of a wide range of functional materials, including highly conductive metals and electrically insulating ceramics, under ambient conditions.			

	The technology targets inline-capable system concepts suitable for industrial manufacturing environments and enables the digital fabrication of printed electronics, sensors, and functional surfaces.
Rationale for Joint R&D Project	The industrial implementation of LIFT technology requires the combination of laser process physics, thin-film characterization, and system engineering with industrial manufacturing requirements. Fraunhofer ILT provides strong expertise in laser–matter interaction, beam shaping, and diagnostics, while Korean SMEs contribute application-driven requirements, system integration capabilities, and market access. Joint R&D is essential to validate LIFT processes for industrially relevant materials, scale the process toward inline production, and establish commercialization pathways.
Key Words	1. Laser-induced forward transfer (LIFT) 2. Digital thin-film processing 3. Printed electronics 4. Functional surfaces 5. Inline manufacturing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	- Establishment of a robust LIFT process for conductive metal and ceramic insulating films - Development of an inline-capable LIFT system architecture - Demonstration of stable thin-film transfer with defined functional properties - Validation of process scalability toward industrial manufacturing			
R&D Target Specifications	- Electrical conductivity of silver films: > 60% of bulk conductivity - Dielectric strength of ceramic insulating films (e.g., SiN): > 10 kV/mm - Reproducible film thickness and homogeneous morphology - Maskless, digitally controllable film deposition			
R&D Plan	Phase 1 – Process fundamentals and material screening (TRL 4) - Investigation of suitable donor foils and functional materials - Analysis of laser parameters (pulse duration, fluence, beam shape) - Evaluation of transfer stability and resolution Phase 2 – System development and process control - Design of optical system for homogeneous energy distribution - Development of foil handling and positioning concepts - Integration of monitoring and control strategies Phase 3 – Pilot-scale validation (TRL 5–6) - Integration of LIFT module into inline demonstrator system - Functional characterization of transferred films (electrical, dielectric, structural) - Evaluation of repeatability, throughput, and scalability			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer ILT will be responsible for: - Development of laser-based LIFT processes - Optical system design and beam shaping - Thin-film characterization (SEM, AFM, electrical testing, XRD) - Process optimization and pilot-line validation - Provision of background IP for research purposes
Expected R&R of Korean SME Partner	The Korean SME partner will: - Define industrial requirements and target applications - Provide relevant substrates and test samples - Integrate LIFT technology into demonstrator systems - Conduct application-oriented testing and validation - Lead commercialization strategy and market introduction

5. Expected Outcomes/Impact

Expected Outcomes and Impact	- Prototype-level inline-capable LIFT system for metal and ceramic thin films - Demonstration of digital thin-film manufacturing without masks or vacuum systems - Generation of new intellectual property for both partners - Strengthening of Korea's competitiveness in advanced manufacturing technologies - Establishment of follow-up projects toward full commercialization (TRL 8–9) The developed technology contributes to next-generation digital manufacturing, flexible electronics, and functional surface engineering.

6. References

(1) Fraunhofer ILT internal research on laser-based additive manufacturing

(2) Prior feasibility studies on laser-induced forward transfer processes

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization